

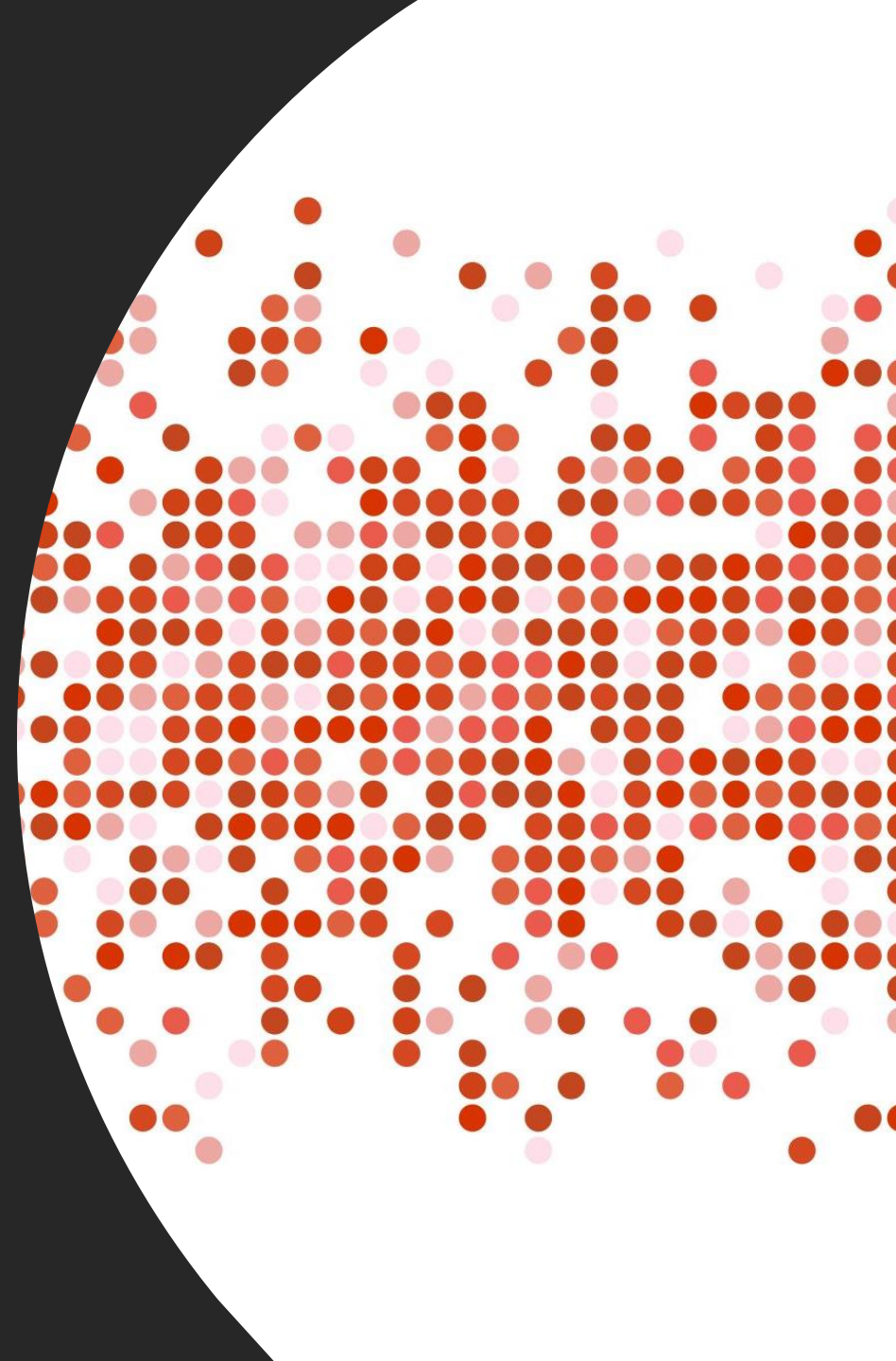
From Policy Confusion to Compliance Transparency:

*Embedding **Quality Assurance & Auditing**
in AI-Assisted Legal Document
Processing Systems*

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Maynooth International Engineering College (MIEC)

June 12th, 2026



Building Responsible Agentic AI Systems

- Understand the Flaws in LLM's Judgement: Structural & Inherited
- How to Measure & Audit LLM's Success in Judgement
- How to Align LLM with Human Principles

The Complexity for Analyzing Digital Trade Policy

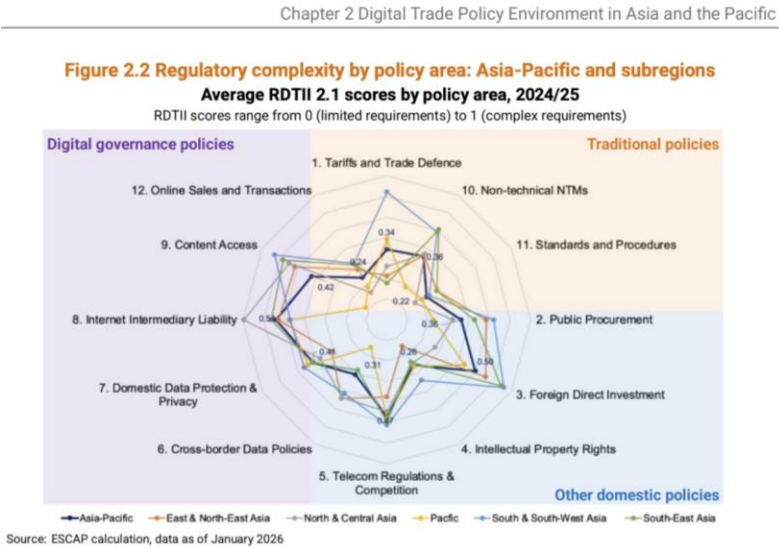
Challenges at a glance



Legal Corpus & Policy Life Cycle



Policy Complexity & Regional Divergence



Standards Alignment

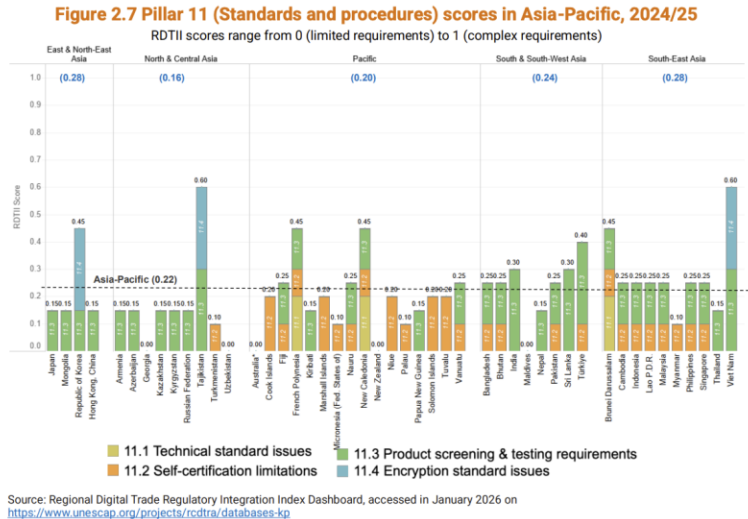


Image Source: AI-generated image, ChatGPT .

Image Source: Digital trade regulatory review for Asia and the Pacific, 2025, <https://repository.unescap.org/server/api/core/bitstreams/a1b649cb-b894-4607-9ef5-154e0502eada/content>

The Searching Interface is Changing, But the Complexity Never Decreases



1

Tools $\neq$ Effectiveness

2

Tools have
(steep/hidden)
learning curves.

3

Tools have flaws.

4

Tools need to be
adapted to be useful.

The Promise of LLMs



Save Time

Speed up time-consuming document-processing tasks, such as summarization, retrieval, reference finding, OCR.



Make Searching More Personal & Context-aware

Investigate each case with content that's tailored to the context.



Navigating Ambiguity

LLM can progress when a situation, or information is incomplete, underspecified, conflicting, or open to multiple interpretations.



Workflow Automation

Agentic AI systems that can plan, make decisions, use tools, and take actions toward a goal with limited human supervision.

However, ...

+

Ask anything



Voice

ChatGPT is AI and can make mistakes.



Hallucination



**Non-
Transparency
(Blackbox)**



**Out of
Alignment**



Costly

Building

Responsible

Agentic AI Systems

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Common **Misunderstanding** About LLM Intelligence

Fill in the blanks with the correct word.

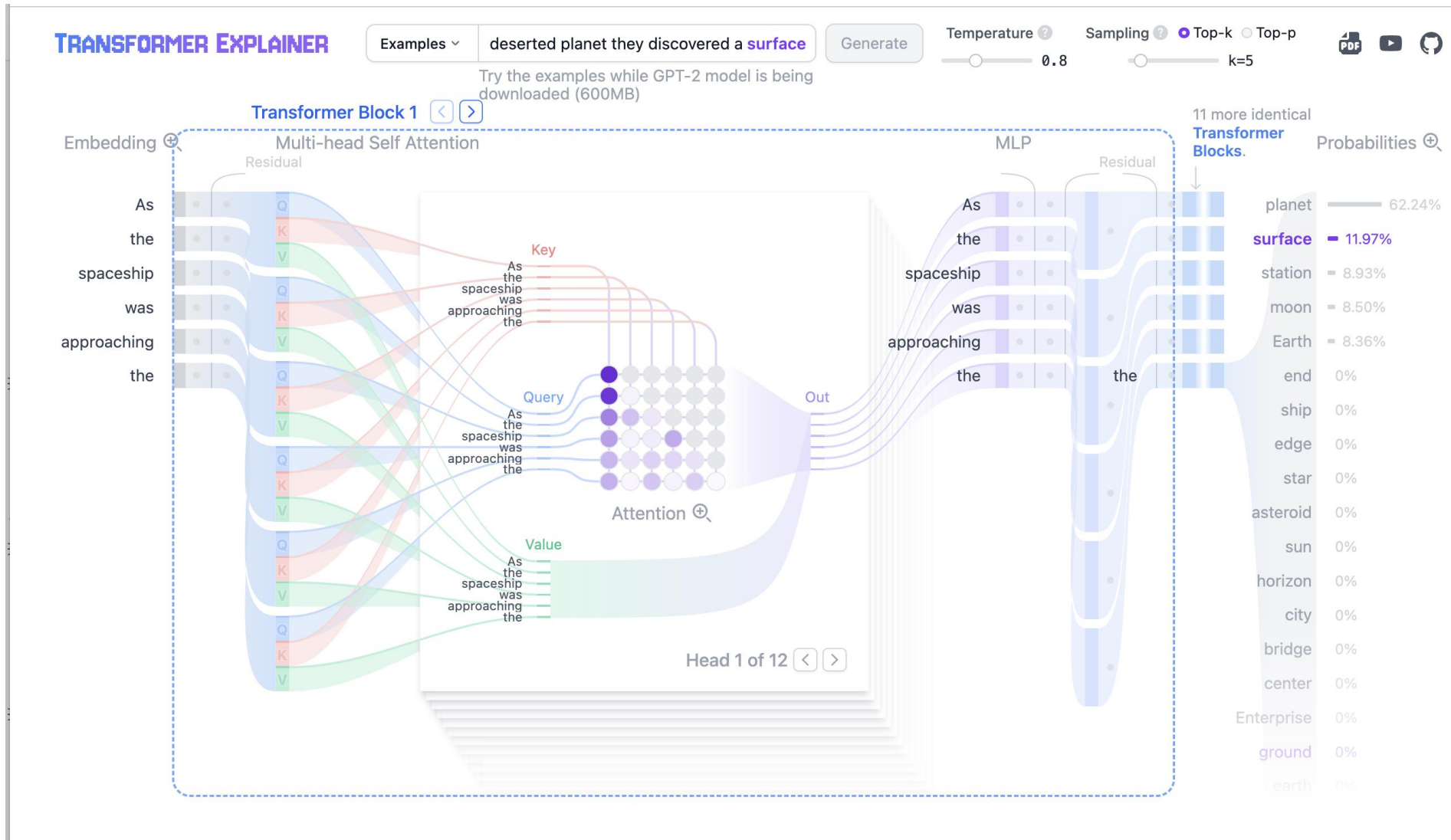
The
Filling-In-Blanks Game
≠
Reasoning

sun	cap	bed	run	log
map	net	pig	cup	jam

- 1.The ____ shines brightly in the sky.
- 2.I wear a ____ on my head.
- 3.I sleep in my ____ at night.
- 4.We like to ____ around the playground.
- 5.The frog sat on a ____ in the pond.
- 6.Use a ____ to find your way.
- 7.The fisherman caught fish with a ____.
- 8.The little ____ rolled in the mud.
- 9.Please pour milk into the ____.
- 10.I spread strawberry ____ on my toast.

LLM is ANOTHER form of intelligence: Its foundation is **statistical**.

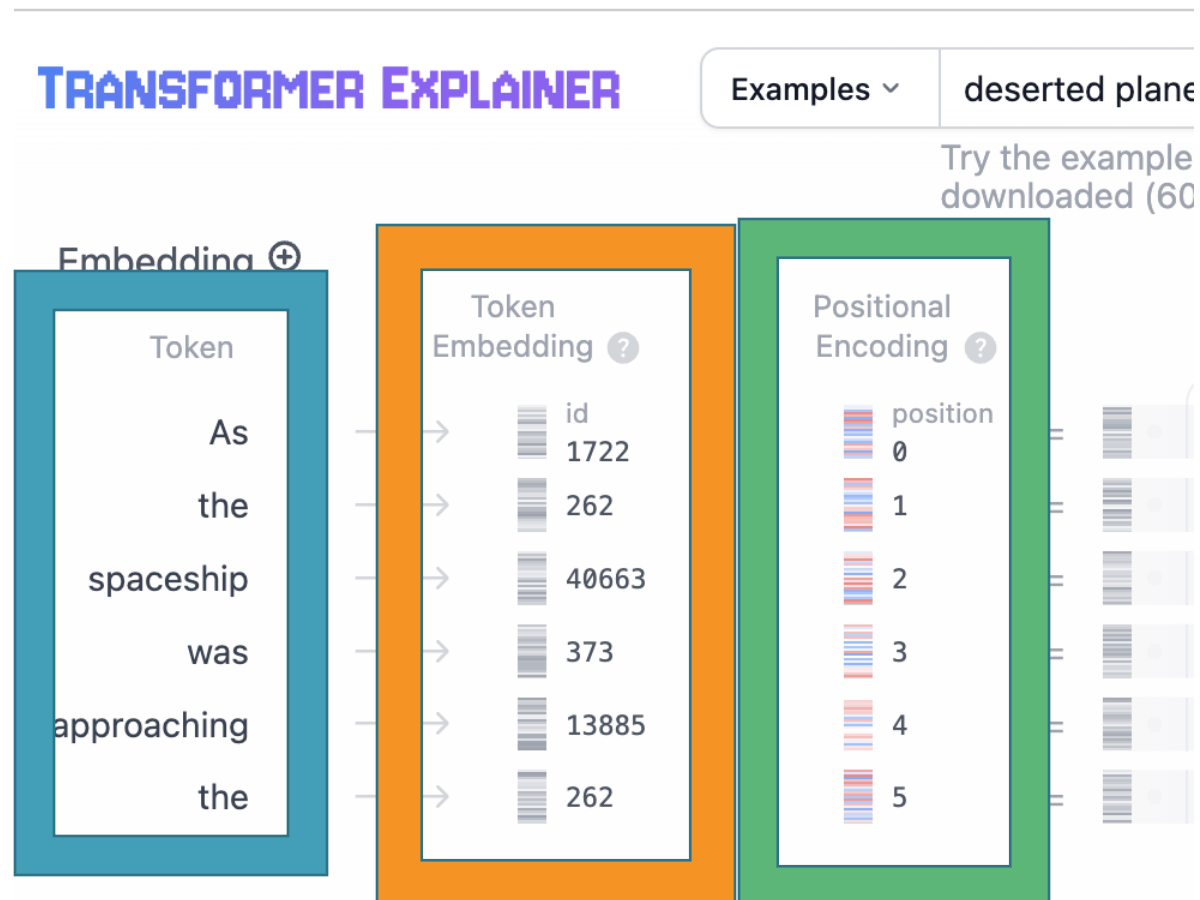
Transformer



Source: Transformer Explainer, <https://poloclub.github.io/transformer-explainer/>

Inside Transformer

- Tokens
- Embeddings

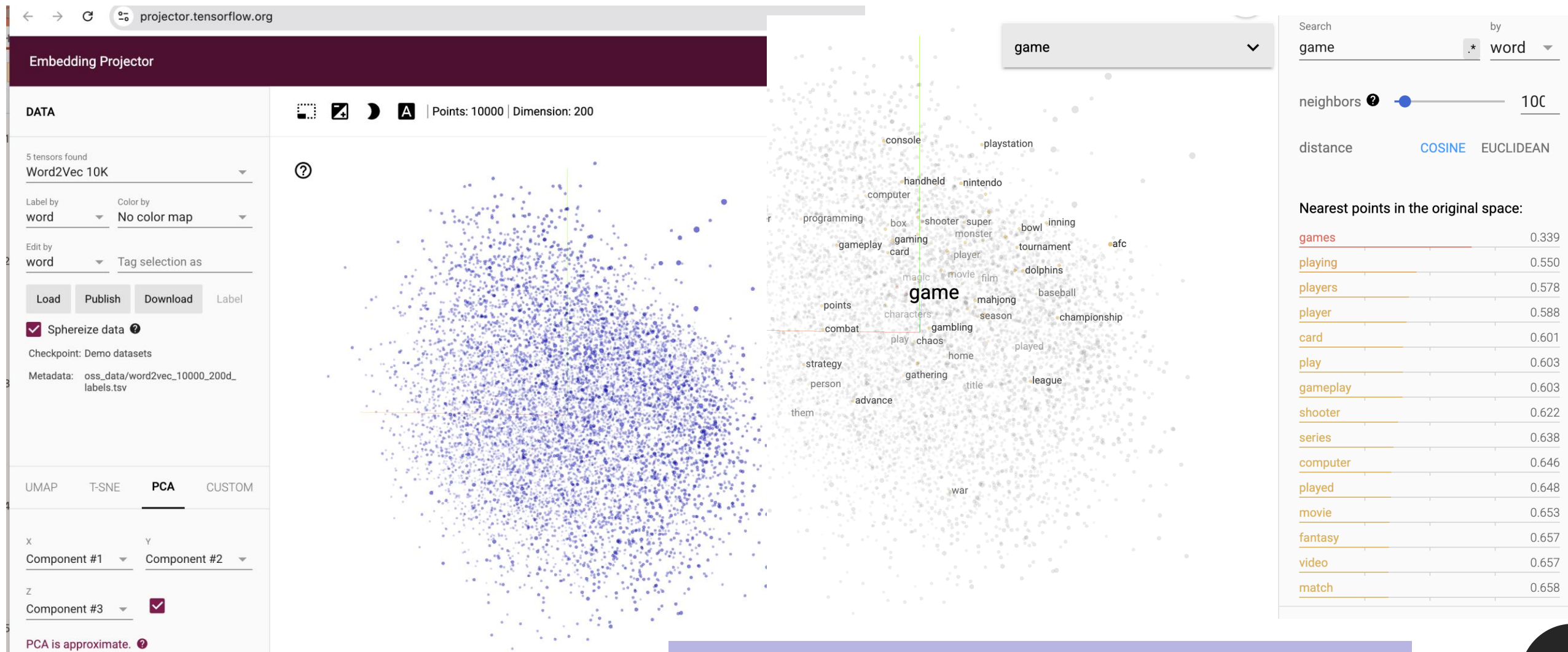


Token Embedding:

In GPT-2 (small),

- Each token in the vocabulary as a **768**-dimensional vector.
- **50,257** tokens in GPT-2's vocabulary.
- These embedding vectors are stored in a matrix of shape **(50257 × 768)**.

Word Embeddings

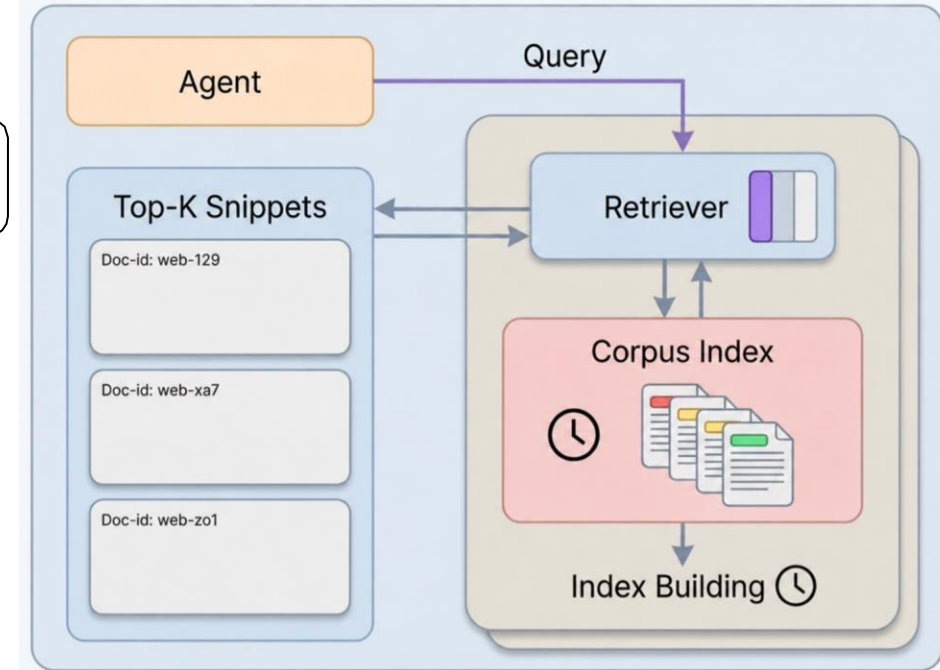
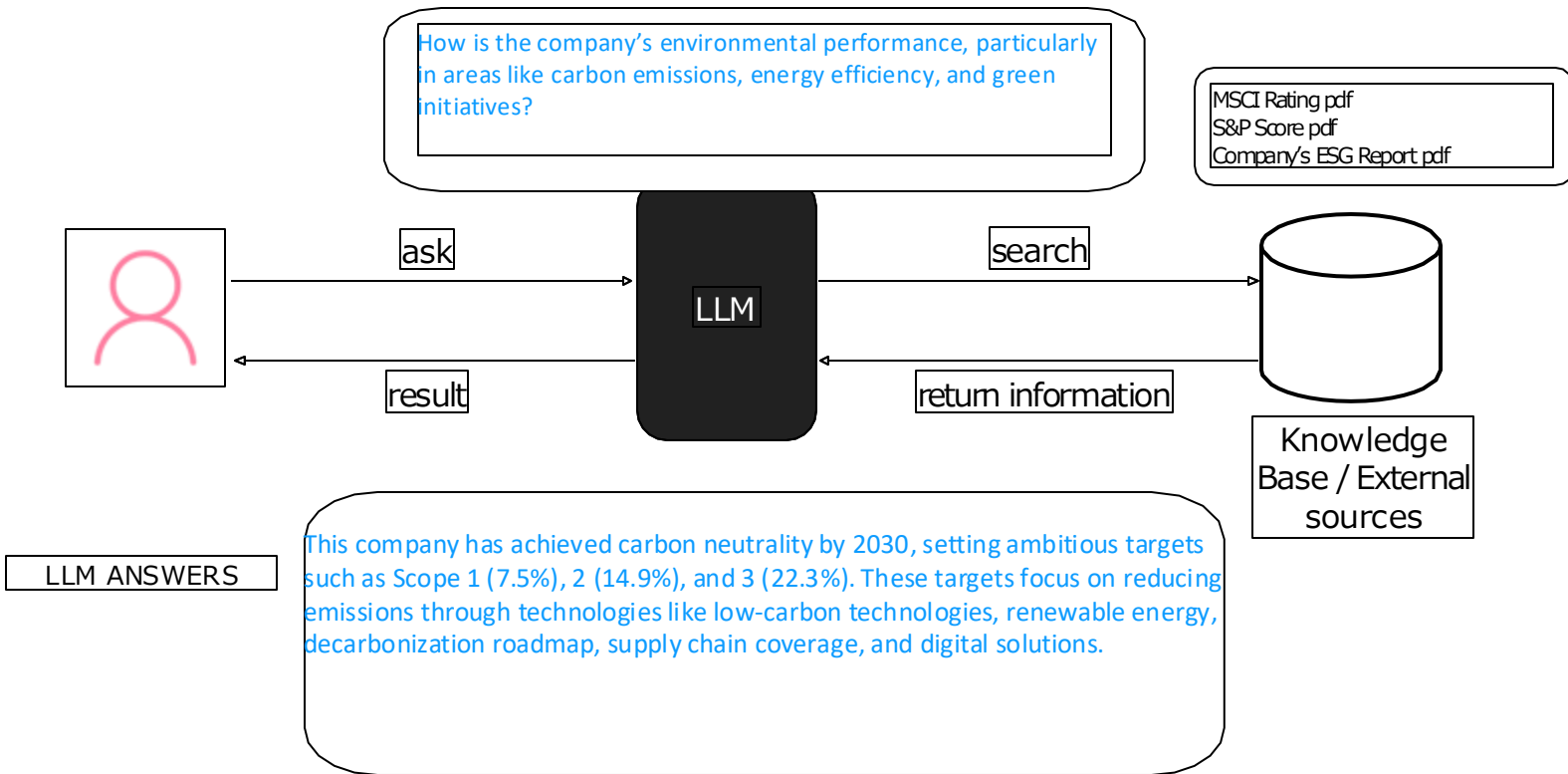


Nearest points in the original space for the word "game"

Source: <https://projector.tensorflow.org/>

Extract Information from Documents (**RAG**)

Retrieve information to **A**ugment the model's knowledge and **G**enerate the output



Example adapted from OpenAI's Cookbook Example Slides:

https://github.com/openai/openai-cookbook/blob/main/examples/data/example_pdfs/rag-deck.pdf

Extract Information from Documents (RAG)



Data preparation

- Chunking
- Embeddings
- Augmenting content



Input processing

- Input augmentation
- Embeddings



Retrieval

- Search
- Multi-step retrieval
- Re-ranking



Answer Generation

- Context window
- Chat Completion

Source: OpenAI Cookbook Example, https://github.com/openai/openai-cookbook/blob/main/examples/data/example_pdfs/rag-deck.pdf



Data preparation: augmenting content

Augmenting content refers to modifications of the original content to make it more digestible for a system relying on RAG. The modifications could be a change in format, wording, or adding descriptive content such as summaries or keywords.

* LLM can do this for you with the right prompt

BEST PRACTICES

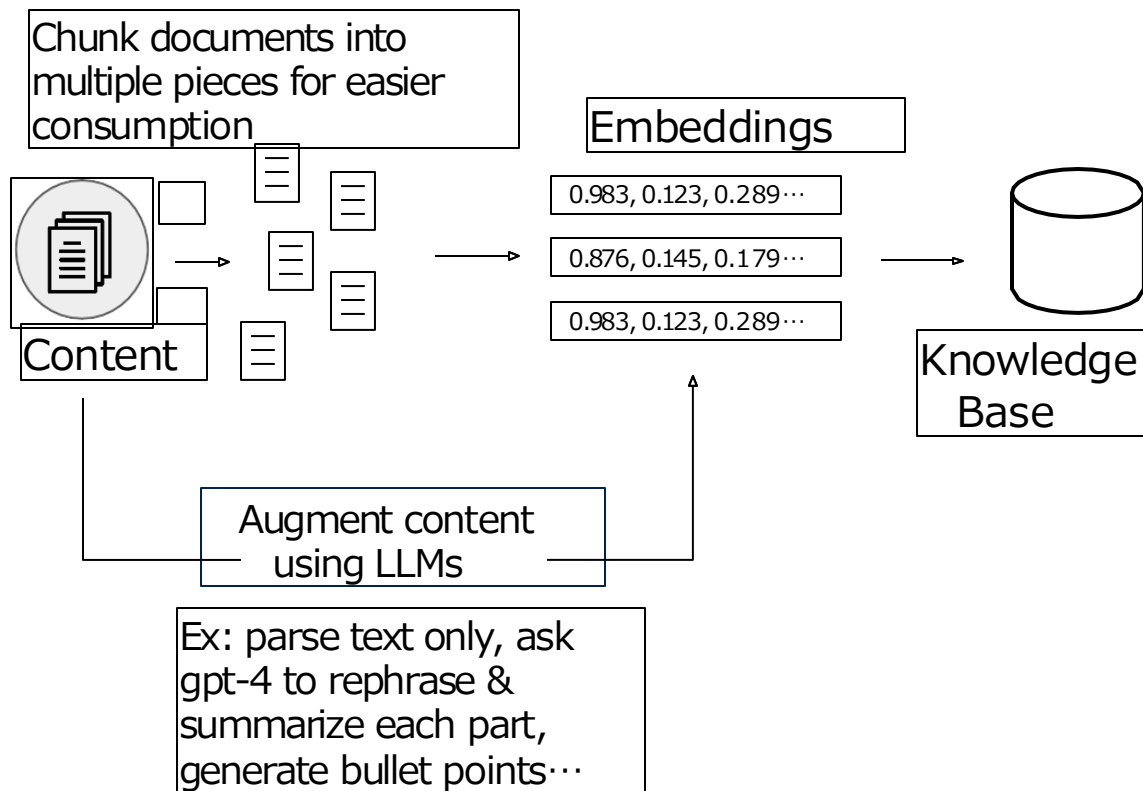
Pre-process content for LLM consumption:
Add summary, headers for each part, etc.
+ curate relevant data sources

COMMON PITFALLS

- Having too much low-quality content
- Having too large documents



Input processing



BEST PRACTICES

- Consider how to transform the input to match content in the database.
- Consider using metadata to augment the user input.

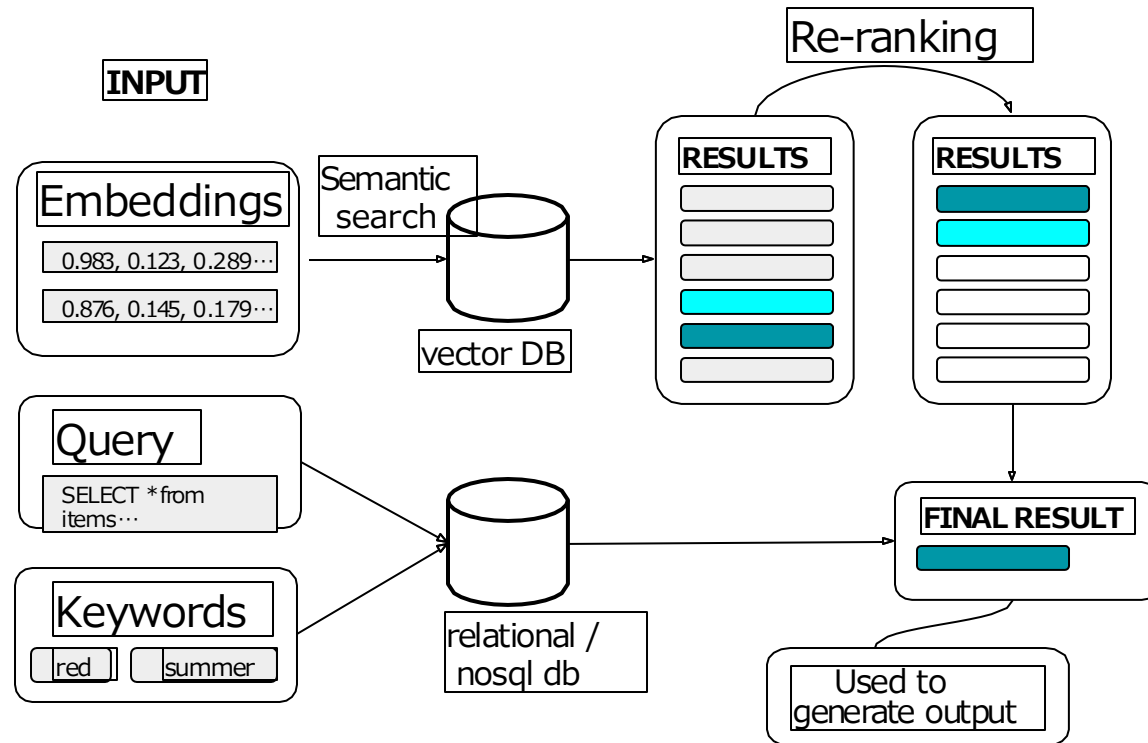
COMMON PITFALLS

- Comparing directly the input to the database without considering the task specificities

Source: OpenAI Cookbook Example, https://github.com/openai/openai-cookbook/blob/main/examples/data/example_pdfs/rag-deck.pdf



Retrieval



BEST PRACTICES

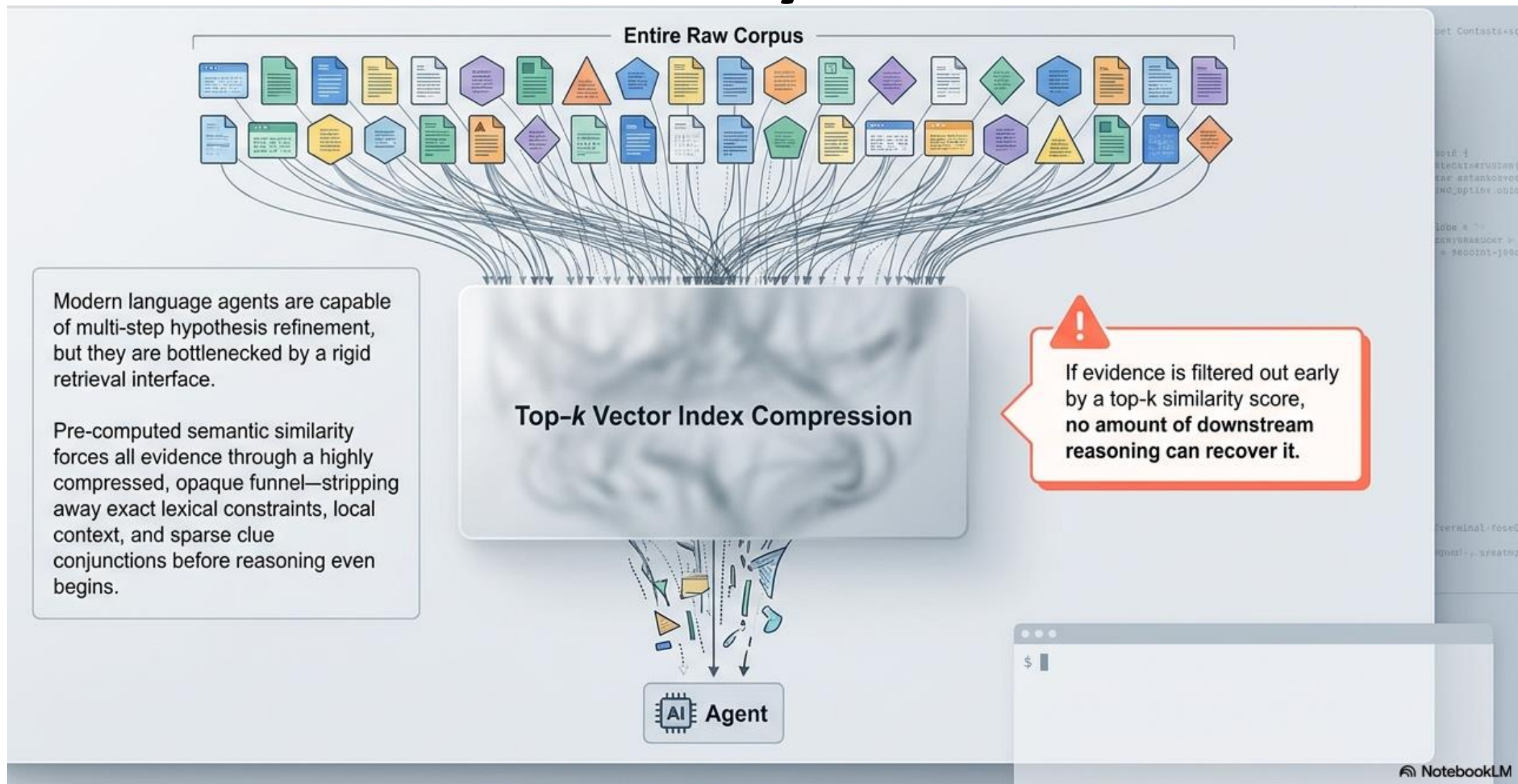
Use a combination of semantic search and deterministic queries where possible

+ Cache output where possible

COMMON PITFALLS

→ The wrong elements could be compared when looking at text similarity, that is why re-ranking is important

The Semantic Similarity Bottleneck



- Li, Zhuofeng, et al. "Beyond semantic similarity: Rethinking retrieval for agentic search via direct corpus interaction." arXiv preprint arXiv:2605.05242 (2026).

Recent Findings: Direct Corpus Interaction

- Modern retrieval systems expose a corpus through a fixed similarity-based interface.
- They typically compress access into a single **top-k retrieval** step before reasoning begins. This approach is efficient, but it can become a bottleneck for agentic search.
- Conventional retrievers struggle with exact lexical constraints, sparse clue combinations, local context checks, and multi-step hypothesis refinement.
- Evidence filtered out during early retrieval cannot be recovered later, even with stronger downstream reasoning.

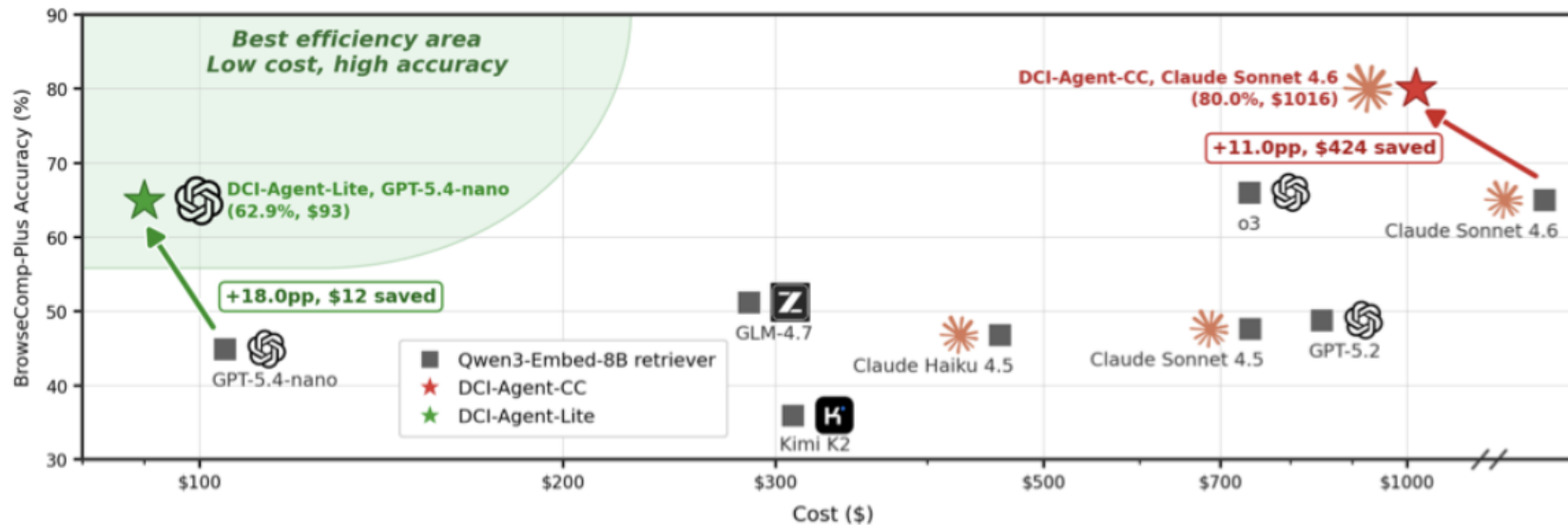
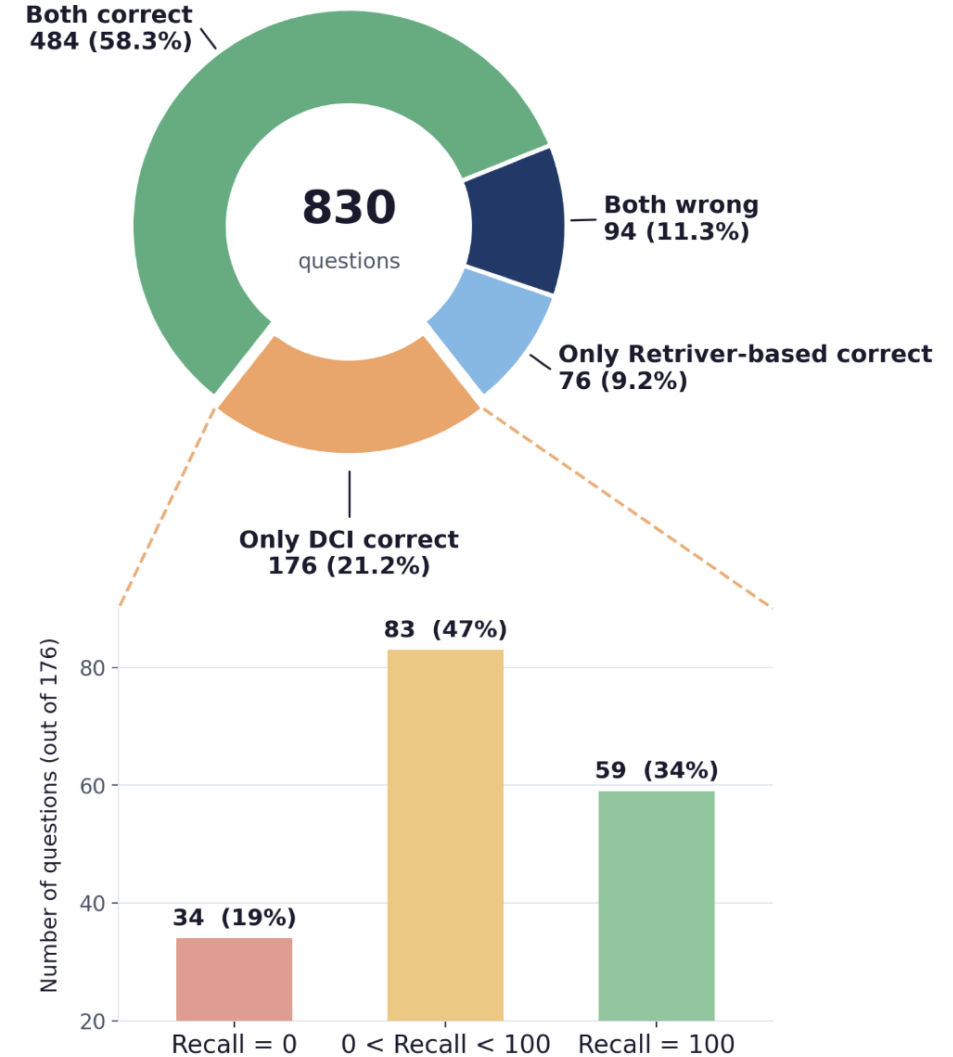


Figure 1: Pareto frontier of performance vs. cost on BrowseComp-Plus, comparing two paradigms: *retrieval* (Qwen3-Embed-8B retriever) and our proposed *direct corpus interaction* (two DCI-Agents ★ and ★). The x-axis shows the estimated API cost over the entire evaluation set.

- Li, Zhuofeng, et al. "Beyond semantic similarity: Rethinking retrieval for agentic search via direct corpus interaction." arXiv preprint arXiv:2605.05242 (2026).

Closing the Gap on post-retrieval failures

- DCI-Agent-CC correctly answers 176 BrowseComp-Plus questions that the matched retrieval agent misses, whereas only 76 show the reverse pattern.
- Among the 176 CC-win cases, only 34 contain no gold documents retrieved by the retrieval agent.
- The vast majority of DCI's wins occurred when RAG only surfaced the evidence, but the agent couldn't bridge the hops or constraints using limited snippets retrieved by top-k similarity scores.

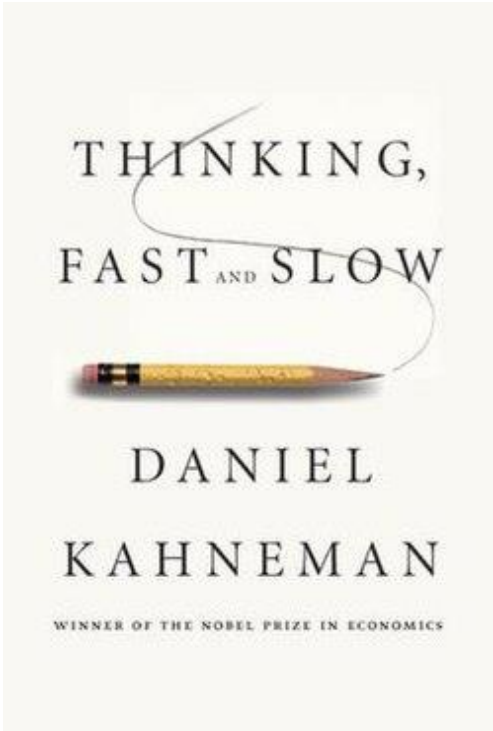


- Li, Zhuofeng, et al. "Beyond semantic similarity: Rethinking retrieval for agentic search via direct corpus interaction." arXiv preprint arXiv:2605.05242 (2026).

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AI Agent: Build System 2.0 into your LLM

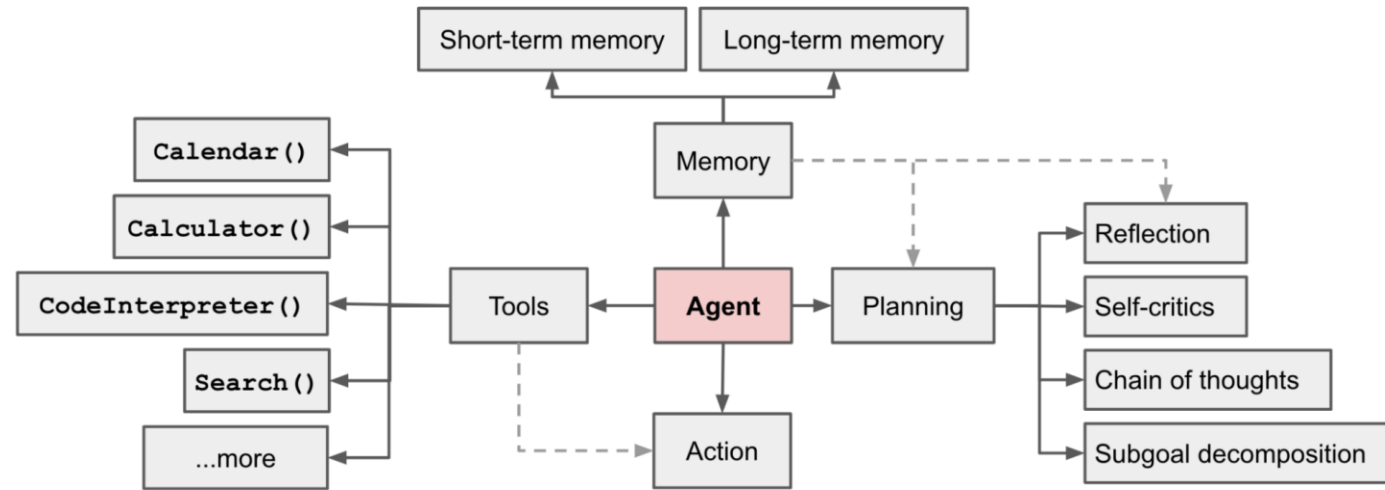


The Two Systems of Thinking:

- System 1 (Fast Thinking):** Intuitive, automatic, emotional, and effortless. It makes quick judgments based on patterns and past experiences.
- System 2 (Slow Thinking):** Deliberate, logical, and effortful. It is used for complex reasoning, problem-solving, and critical thinking.

"Thinking, Fast and Slow" by **Daniel Kahneman** is a book that explores how human thinking is shaped by two distinct cognitive systems. Kahneman, a Nobel Prize-winning psychologist, explains the biases and errors in decision-making through research in behavioral economics and psychology.

Source: https://en.wikipedia.org/wiki/Thinking,_Fast_and_Slow

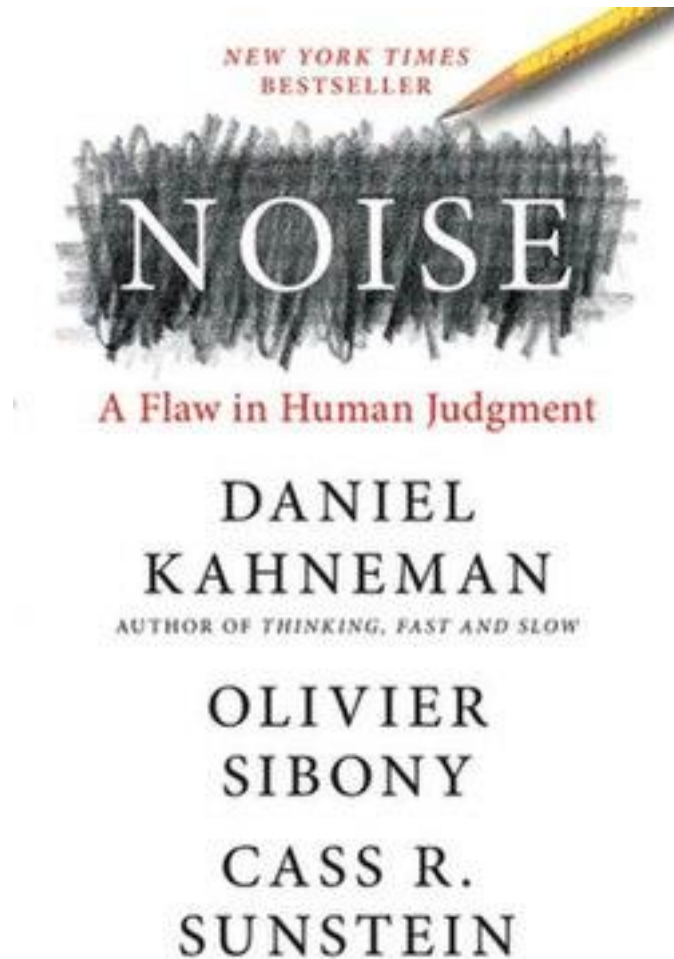


LLM Powered Autonomous Agents:

<https://lilianweng.github.io/posts/2023-06-23-agent/>
Lilian Weng

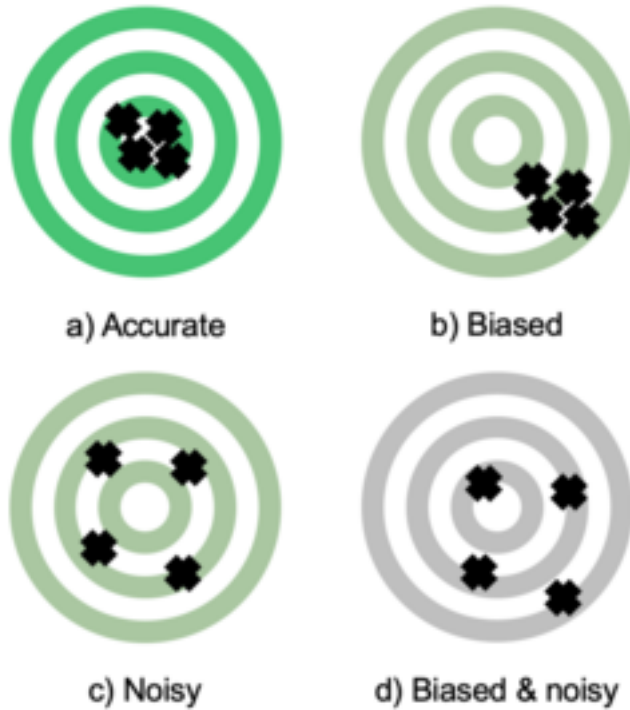
Co-founder of Thinking Machines Lab [@thinkymachines](https://thinkymachines.com/) ;
Ex-VP, AI Safety & robotics, applied research [@OpenAI](https://openai.com/)

A Flaw in Human Judgement



- Kahneman, Daniel, Olivier Sibony, and Cass R. Sunstein. *Noise: A flaw in human judgment*. Hachette UK, 2021.
- ***Noise: A Flaw in Human Judgment*** is a nonfiction book (2021) by professors Daniel Kahneman, Olivier Sibony and Cass Sunstein.
- Undesirable variability in judgments of the same problem
- The statistical properties and psychological perspectives of this issue

A Flaw in Human Judgement



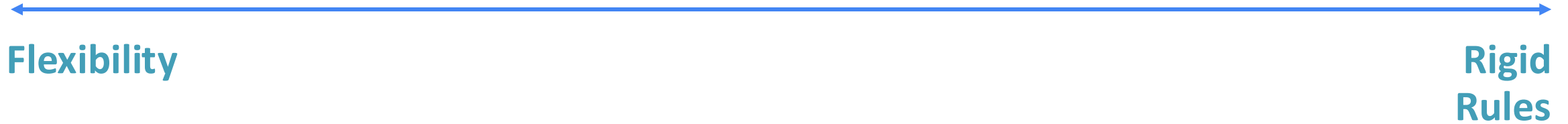
Criminal law

A study on 208 criminal judges showed that their independently given punitive recommendations on 16 fictive cases varied greatly in harshness. For example, the judges only unanimously recommended imprisonment in three of the cases, and while the recommended number of prison years in one case was 1.1 years on average, one recommendation was as high as 15 years.

Noise detection → difficult

However, ...

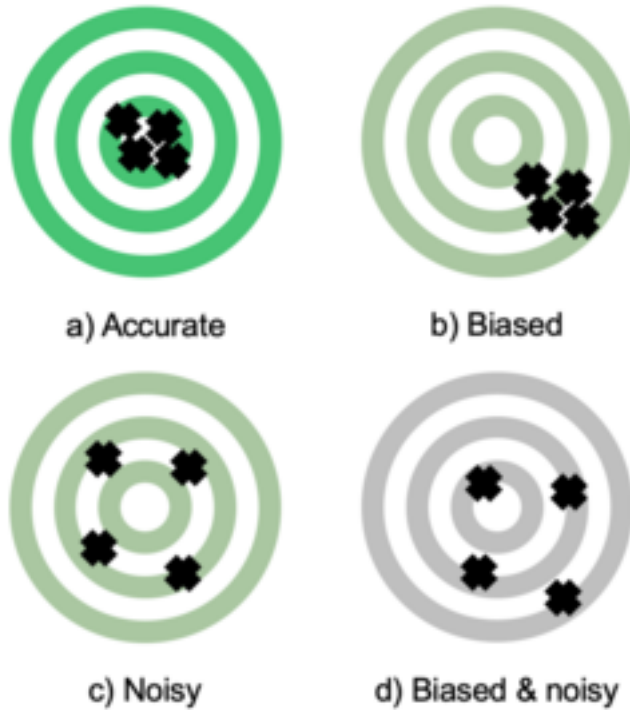
Reaching zero noise may not be desirable since it would reduce flexibility.



Whenever there is human judgment, there is Noise.

so it is with LLMs.

The Inherited Flaw of LLM's Judgement



- The wrong question: How accurate can an LLM's judgment be?
- We cannot reliably measure the accuracy of an LLM's judgment, but we can measure its fidelity.

Bad news: For LLM, any single score (numerical number) is not reliable.

Good news: For LLM, simulating ambiguity is computationally cheap.

The Fix: Noise Audit

Bad news: For LLM, any single score (numerical number) is not reliable.

Good news: For LLM, simulating ambiguity is computationally cheap.

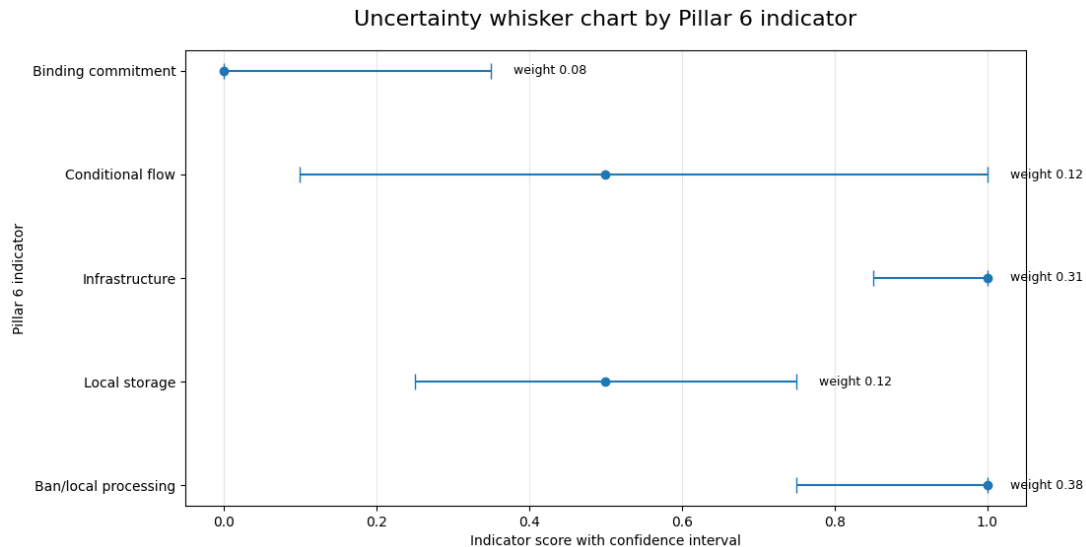


Image Source: AI-generated image, ChatGPT .

In **Noise: A Flaw in Human Judgment**, a **noise audit** is a way to measure how much unwanted variability exists in human judgments.

1. Choose a judgment task.

2. Prepare multiple LLM judges, each with a different persona.

3. Collect LLM judges' conclusions independently

This is important because the audit is measuring individual judgment variation, not group consensus.

4. Measure disagreement among LLM judges

The audit examines how much judgments vary when they should, in principle, be similar. Large differences reveal **noise**.

5. Compare actual disagreement with expected disagreement.

Use statistical measures such as Krippendorff's Alpha Values.

The Fix: Noise Audit

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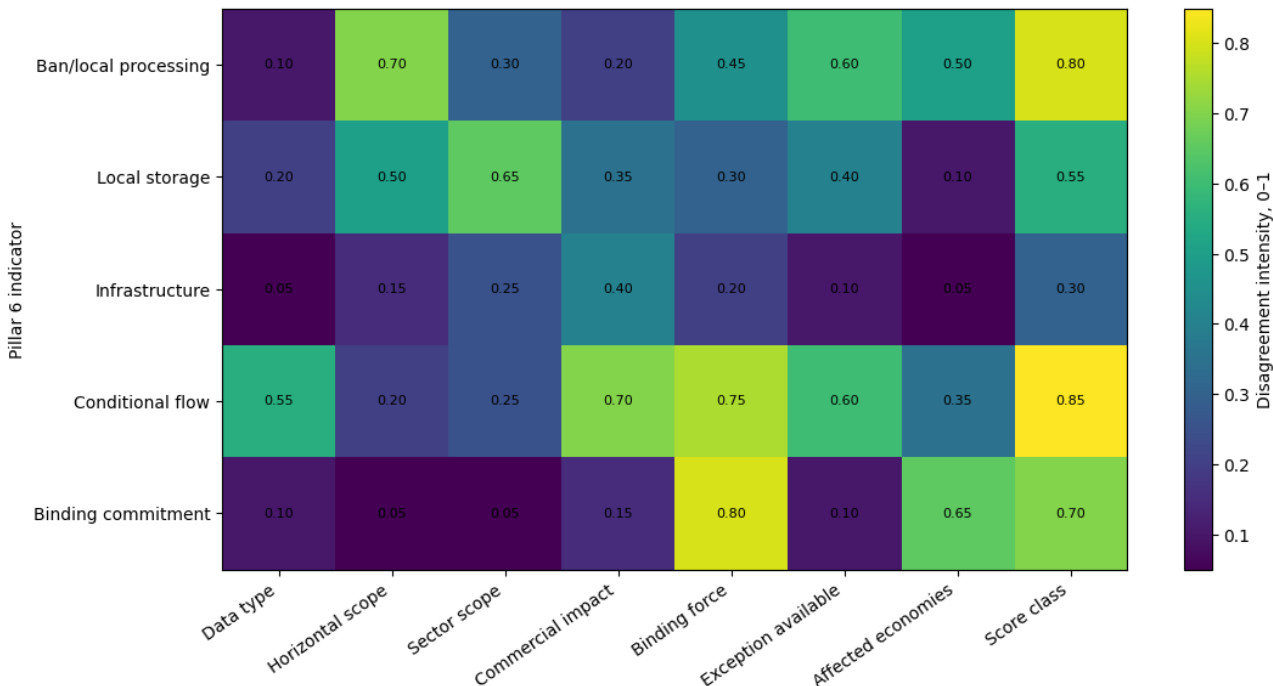


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Use statistical measures such as Krippendorff's Alpha Values.

Tips

Do not collapse evidence into scores too early.

Keep source fragments, interpretations, and scoring decisions separate.

Always model uncertainty.

Legal text is often vague, translated, outdated, or sector-specific.



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Evaluation of LLM's Judgement in Law

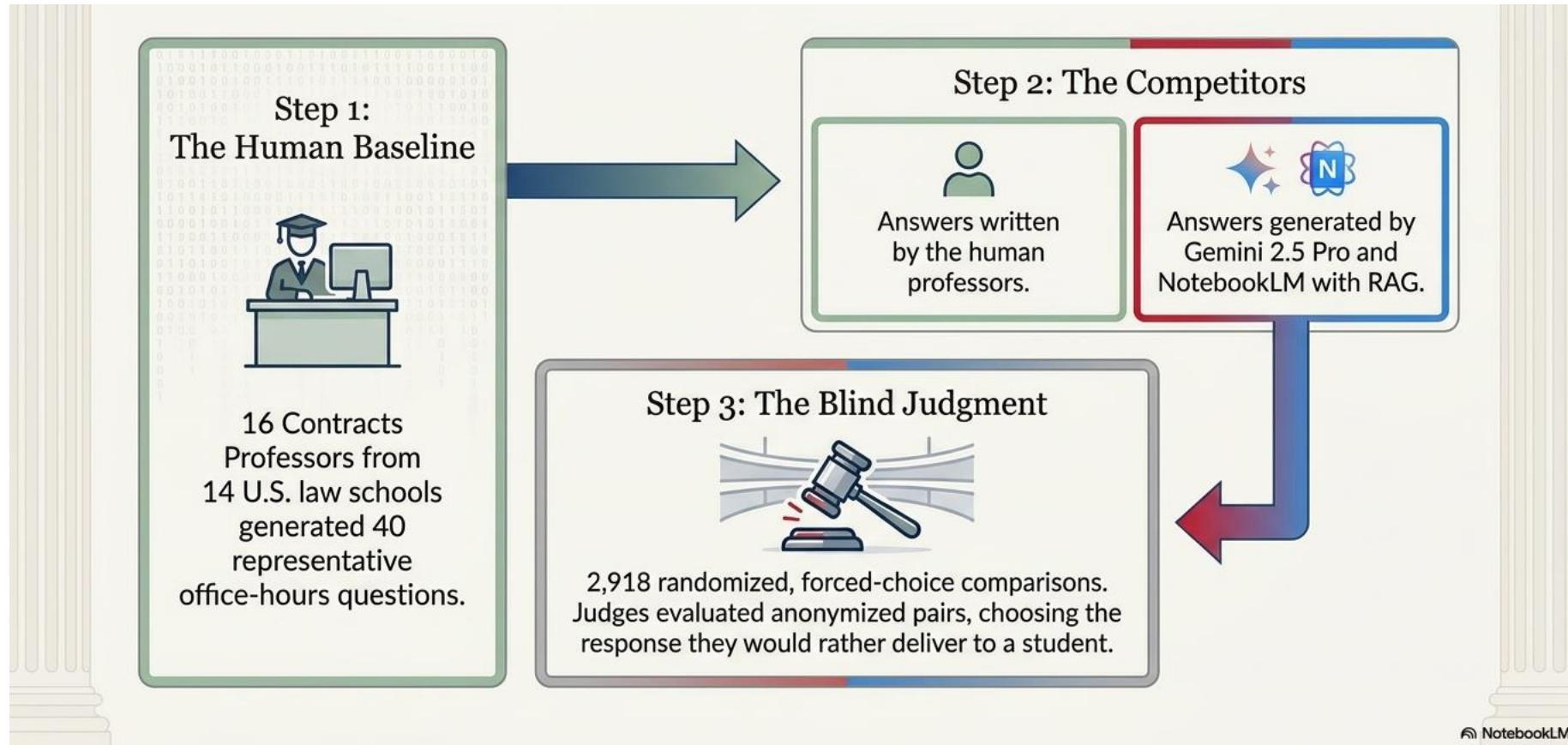
Media: <https://law.stanford.edu/press/ai-outperforms-law-professors-in-stanford-law-study/>



- law, correctness cannot be reduced to one solution.
- The task is not just to convey facts, but to cultivate judgment
- The exercise of judgement:
 - weighing competing reasons
 - grappling with ambiguity
 - reaching a defensible conclusion

- Salinas, Alejandro, et al. "Law Professors Prefer AI Over Peer Answers." (2026).

Evaluation of LLM's Judgement in Law



- Salinas, Alejandro, et al. "Law Professors Prefer AI Over Peer Answers." (2026).

Results

When considering the average win rate, LLMs far outperformed human instructors, with Gemini achieving 75.92% across all trials against instructors, while NotebookLM won 74.75%.

- Salinas, Alejandro, et al. "Law Professors Prefer AI Over Peer Answers." (2026).

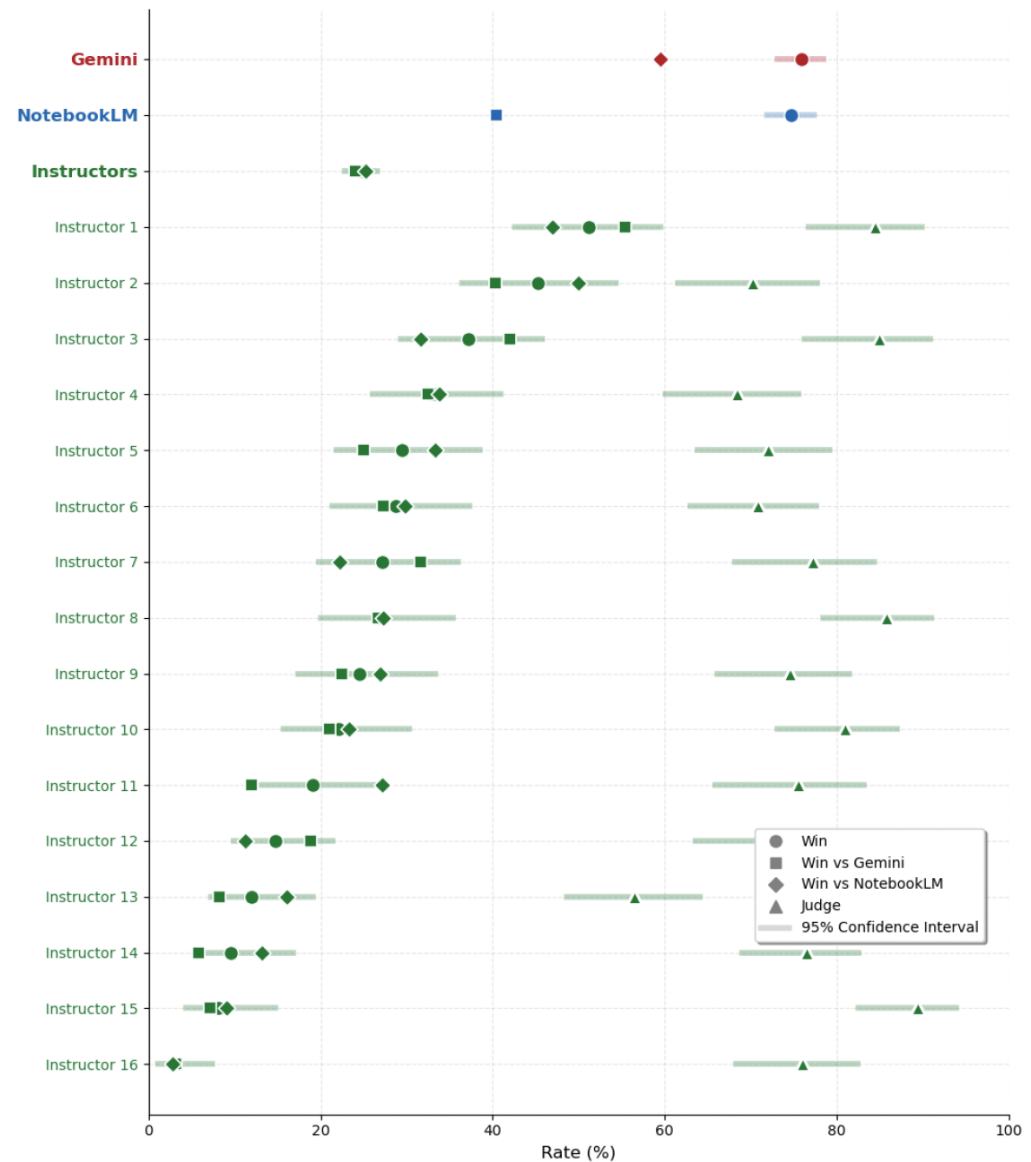


Figure 2: Win rates of instructors compared to LLMs (Gemini and NotebookLM). Circles indicate win rates with 95% Wilson score confidence intervals; diamonds and squares represent win rates against NotebookLM and Gemini, respectively. Triangles indicate LLM-preference rates at the judge level. Models are shown at the top, followed by individual instructors.

Uncertainty of the Win Rates

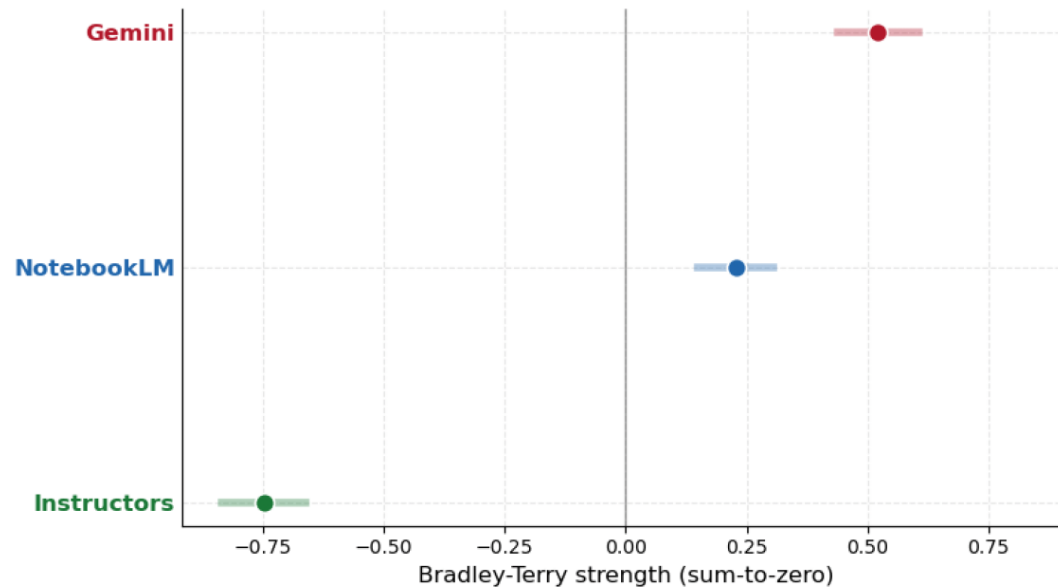
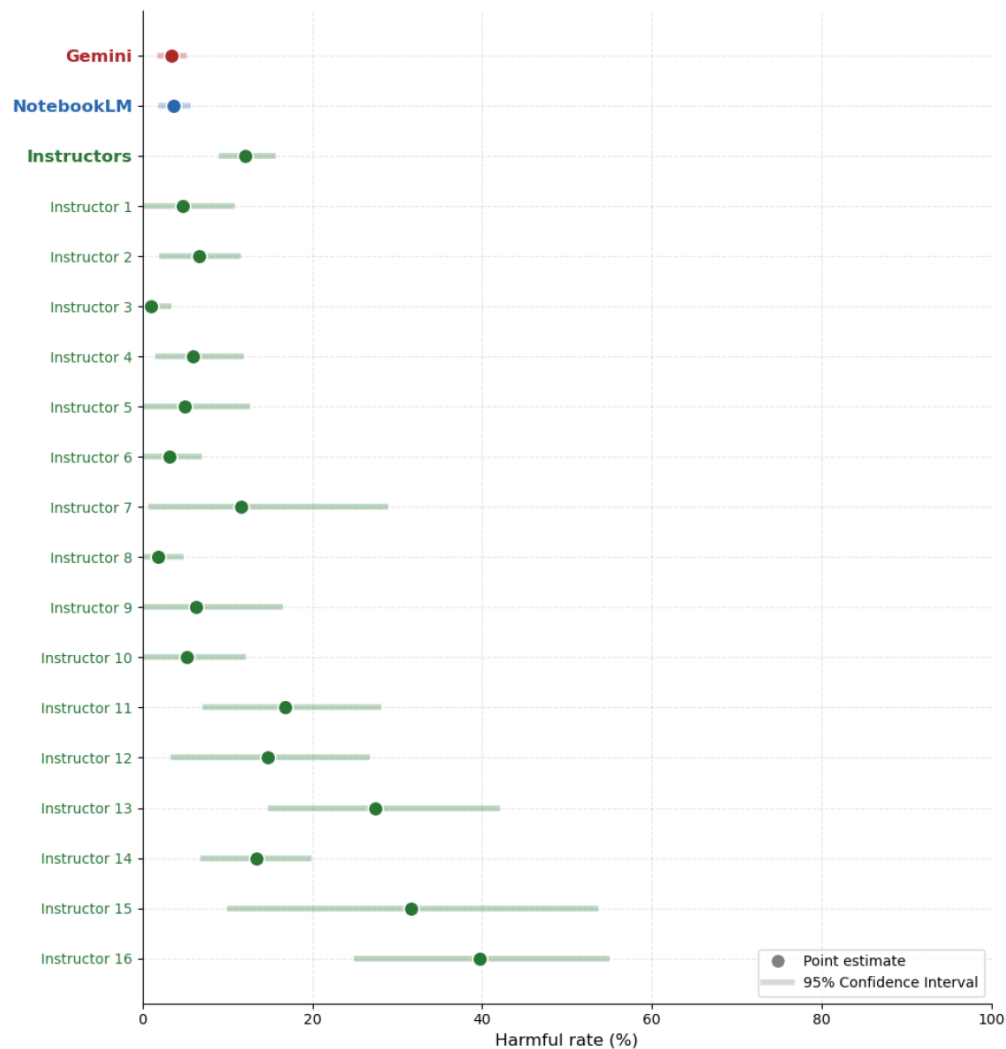


Figure D.1: Bradley-Terry (BT) log-odds strength estimates for Gemini, NotebookLM, and the instructor average. Points represent centered BT coefficients, interpreted as latent performance strengths in pairwise comparisons. Error bars show simultaneous multiplicity-corrected 95% intervals derived from projecting the χ^2 confidence ellipsoid for the full coefficient vector. Higher values indicate stronger estimated performance under the BT model.

- Salinas, Alejandro, et al. "Law Professors Prefer AI Over Peer Answers." (2026).



Harmfulness Rate

Figure F.1: Harmfulness rates for instructors and LLMs. Points denote mean harmfulness rates as defined in Supplementary Information [F](#), with non-parametric 95% bootstrap confidence intervals obtained by resampling responses.

- Salinas, Alejandro, et al. "Law Professors Prefer AI Over Peer Answers." (2026).

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- Liu, Jiacheng, et al. "Dive into Claude Code: The Design Space of Today's and Future AI Agent Systems." *arXiv preprint arXiv:2604.14228* (2026).

The 1.6% Illusion

As AI transitions from predictive autocomplete to autonomous judgment, the LLM itself becomes a tiny fraction of the system. The real engineering marvel is the operational infrastructure that bounds, monitors, and safely executes that judgment.

Only 1.6% of Claude Code's codebase is AI decision logic. The other 98.4% is deterministic infrastructure -- permission gates, context management, tool routing, and recovery logic.

Deterministic Execution Harness
98.4% Operational Infrastructure.

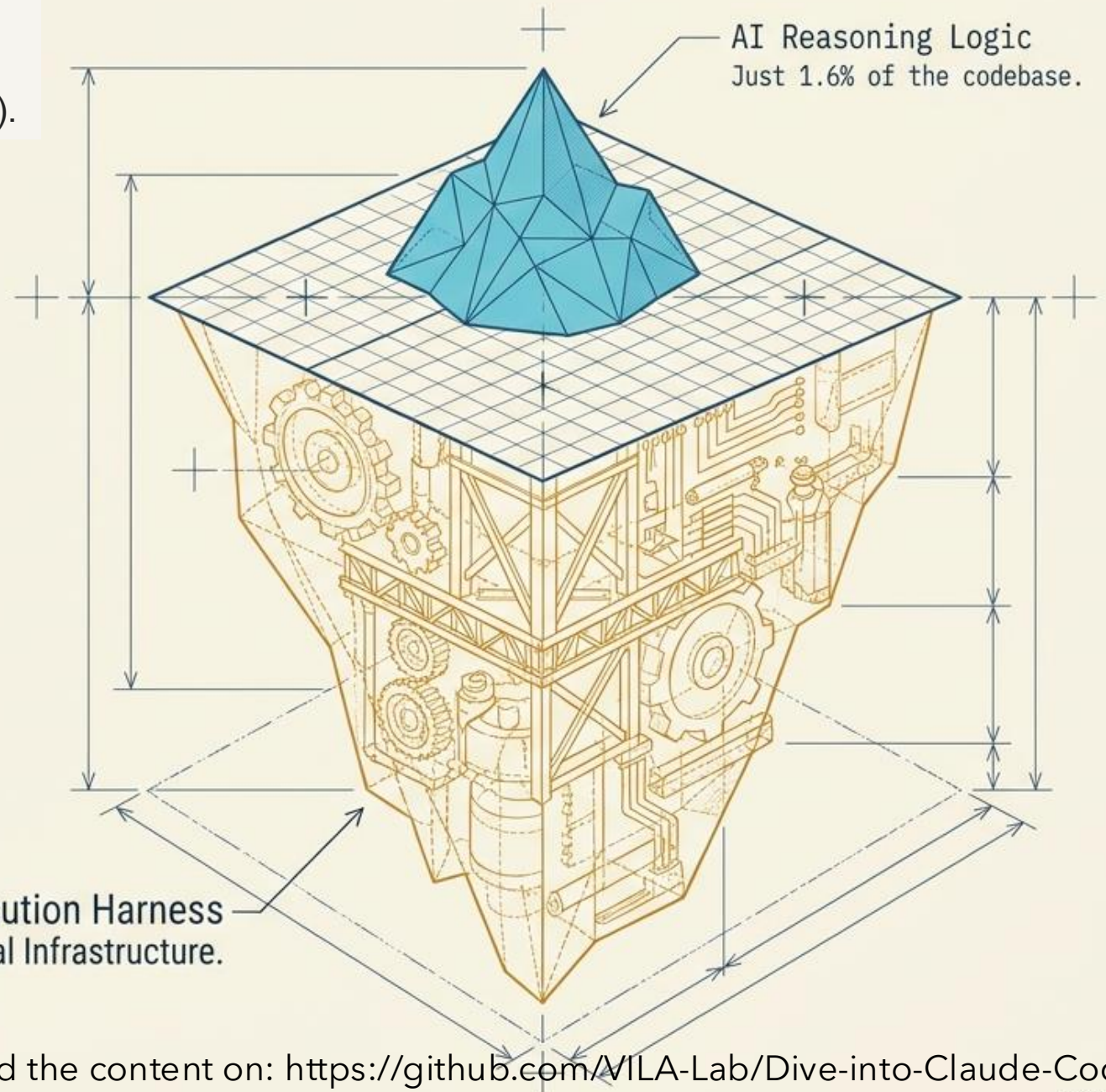


Image was AI-generated and based on the paper and the content on: <https://github.com/VILA-Lab/Dive-into-Claude-Code>

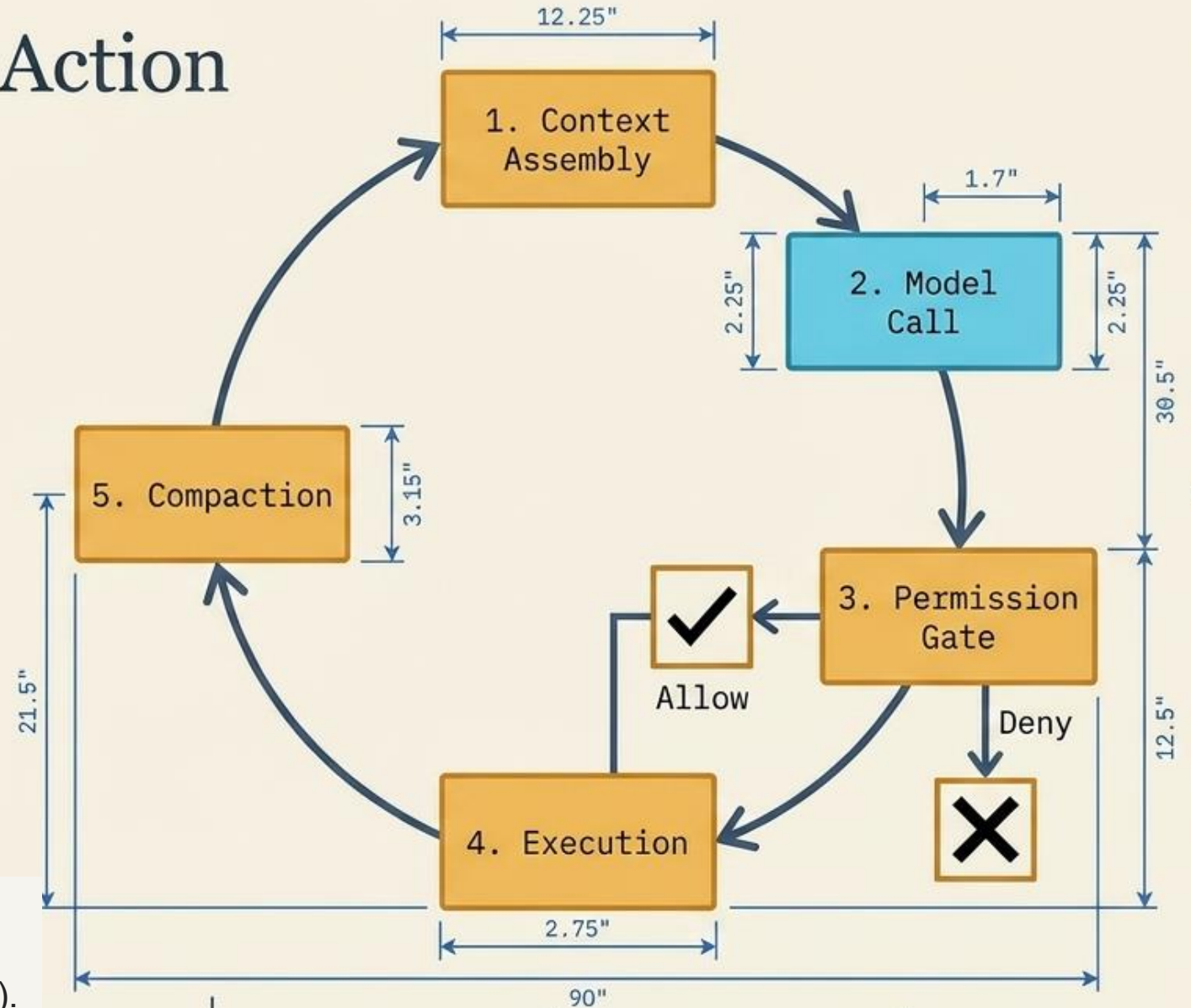
The Agentic Loop in Action

Insight:

Ground truth from the environment assesses progress at every step.

The core is a single while-true cycle. To prevent the model from blindly praising its own mediocre work, generation is strictly separated from evaluation. The loop only advances based on concrete tool returns.

- Liu, Jiacheng, et al. "Dive into Claude Code: The Design Space of Today's and Future AI Agent Systems." *arXiv preprint arXiv:2604.14228* (2026).



Subagent Isolation

Boundaries

To prevent context exposure, subagents operate in isolated windows with restricted tool pools. They return only brief textual summaries to the parent, preserving the primary context budget for high-level judgment.

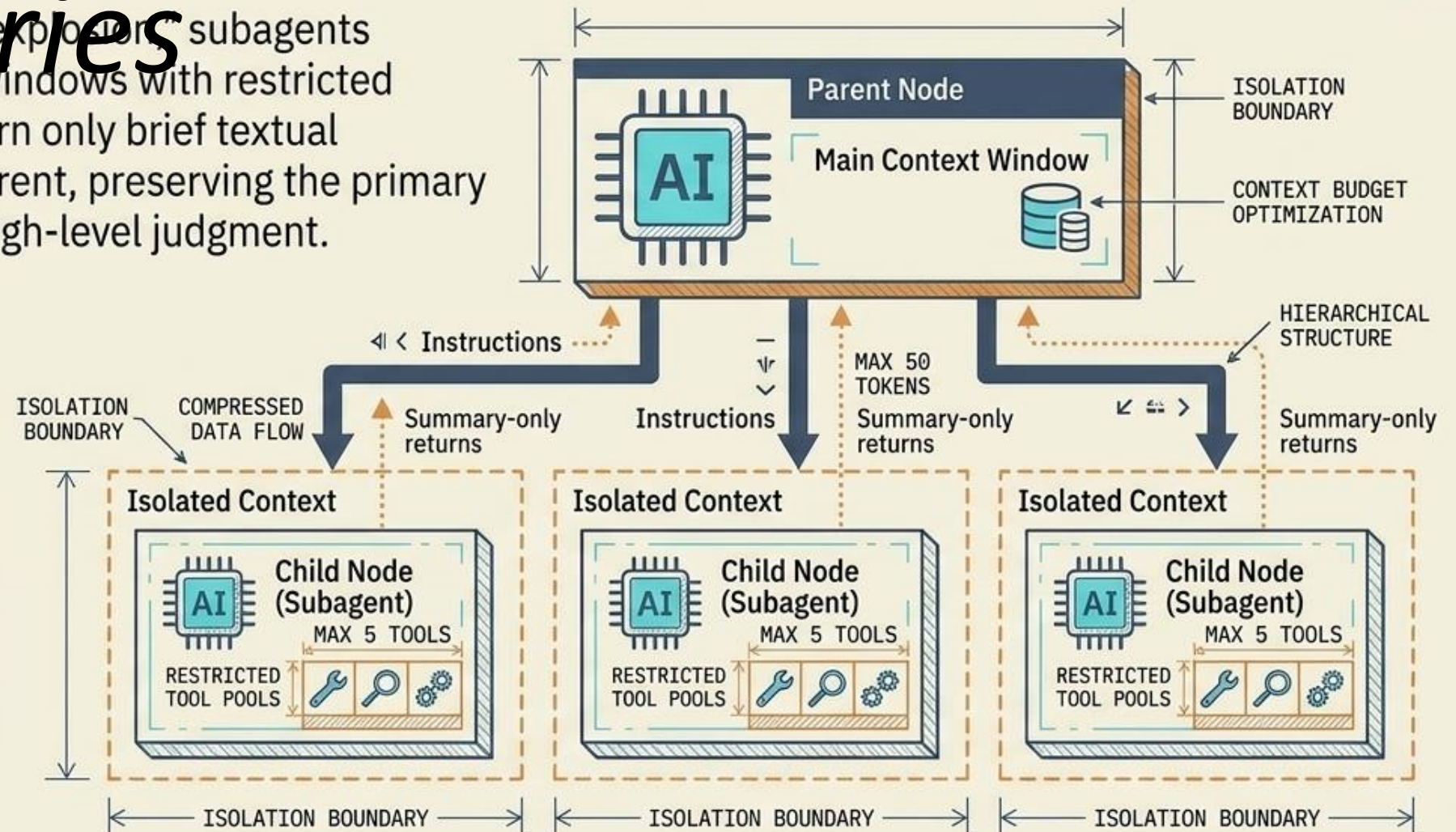


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Anthropic/Claude-for-Legal Plugins

[claude-for-legal](#) / [ai-governance-legal](#) / [skills](#) / [reg-gap-analysis](#) / SKILL.md

markpikemarkpike Remove Lexis references at partner request 210fd7a · last month History

PreviewCodeBlame

241 lines (168 loc) · 14.4 KB

Raw

name	reg-gap-analysis
description	Diff a new AI regulation or guidance against your current governance posture — surfaces gaps, priorities, and a remediation plan with owners and deadlines. Use when an AI regulation moves (or you learn about one you missed), or when user says "new reg just dropped", "does [regulation] affect us", "gap analysis for EU AI Act", "compliance check against [AI law or guidance]", or pastes regulatory text.
argument-hint	[regulation name, or paste regulatory text, or attach a document]

Source: <https://github.com/anthropics/claude-for-legal/blob/main/ai-governance-legal/skills/reg-gap-analysis/SKILL.md>

Anthropic/Claude-for-Legal Plugins

Purpose

The EU AI Act goes live. Colorado passes an AI law. The CFPB issues model risk guidance. The FTC publishes an AI enforcement policy. Something moves — and now you need to know what, if anything, you have to change.

This skill diffs the new requirement against your current AI governance posture (per `~/.claude/plugins/config/claude-for-legal/ai-governance-legal/CLAUDE.md` — use case registry, vendor positions, impact assessment practices, and AI policy commitments) and produces a gap list with a remediation plan.

The AI regulatory landscape is moving faster than any other area of law right now.

When a regulation is genuinely ambiguous, say so. Don't paper over uncertainty — legal teams need to know when they're on solid ground versus when they're making a judgment call.

Source: <https://github.com/anthropics/claude-for-legal/blob/main/ai-governance-legal/skills/reg-gap-analysis/SKILL.md>

Anthropic/Claude-for-Legal Plugins

Workflow

Step 1: Scope the regulation

Before diffing, answer:

- **Does it apply?** Jurisdiction, threshold, sector carve-outs, builder vs. deployer distinction. Research the specific scoping rules in the regulation — don't assume.
- *Builder/deployer matters a lot here.* Many AI regimes impose different obligations on the entity that develops/provides the AI system versus the entity that deploys/uses it. Research which role the company occupies under each regime's definitions. Scope first; don't gap-analyze a law that doesn't apply.
- **When?** Effective date. Enforcement date (often different). Phase-in periods for specific provisions. Verify currency.
- **What's actually new?** Some "new" AI laws largely restate existing legal principles (consumer protection, anti-discrimination, sectoral risk management) applied to AI. Others are genuinely new obligations. **Identify the delta from what you already do, not the full text of the law.**

Source: <https://github.com/anthropics/claude-for-legal/blob/main/ai-governance-legal/skills/reg-gap-analysis/SKILL.md>

Anthropic/Claude-for-Legal Plugins

Research the regulation before building the gap analysis

...

Cite primary sources with pinpoint references. **Flag ambiguity for attorney judgment.**

...

Source attribution tiering. Tag every citation in the gap analysis with its source. For model-knowledge citations, use one of three tiers rather than a single blanket "verify" tag:

- [settled] — stable, well-known statutory and regulatory references unlikely to have changed (e.g., GDPR Art. 22, the existence of Regulation (EU) 2024/1689 as the EU AI Act, Colorado AI Act as C.R.S. § 6-1-1701 et seq.). Still verify before filing, but lower priority.
- [verify] — model-knowledge citations that are real but should be verified: specific delegated / implementing acts, regulator guidance, standards, enforcement actions, case holdings, thresholds, effective dates, phase-in provisions, harmonized-standards references.
- [verify-pinpoint] — pinpoint citations (specific article numbers, annex references, subsection letters, paragraph numbers, standard-clause references) carry the highest fabrication risk and should ALWAYS be verified against a primary source. EU AI Act article numbers in particular shifted during consolidation; every pinpoint cite to the Act should be verified against the Official Journal text.

Source: <https://github.com/anthropics/claude-for-legal/blob/main/ai-governance-legal/skills/reg-gap-analysis/SKILL.md>

Anthropic/Claude-for-Legal Plugins

Output

Save as a dated markdown doc. The remediation plan table becomes a tracker — update status as items close.

If the gap analysis concludes "no gaps, we're compliant," still write the doc. It's useful evidence that you looked, and useful baseline when the regulation is amended.

Cite check before relying on this. Citations here were generated by an AI model and have not been verified against primary sources. Before relying on any citation — statute, regulation, delegated act, guidance, or case — run a verification pass against a legal research tool (Westlaw, CourtListener, or your firm's platform) for accuracy, currency, and subsequent history. Fabricated or misquoted citations in filed materials have resulted in sanctions. Source tags on each citation (e.g., [EUR-Lex], [web search — verify]) show where it came from; verify tags carry higher fabrication risk and should be checked first

-

Source: <https://github.com/anthropics/claude-for-legal/blob/main/ai-governance-legal/skills/reg-gap-analysis/SKILL.md>

Anthropic/Claude-for-Legal Plugins

Research the regulation before building the gap analysis

...

Cite primary sources with pinpoint references. Flag ambiguity for attorney judgment.

...

What this skill does not do

- It doesn't interpret ambiguous regulatory language authoritatively. The EU AI Act in particular has significant interpretive questions that aren't resolved yet. When the reg is genuinely ambiguous: say so, state the conservative read, and flag for outside counsel if the issue is material.

Source: <https://github.com/anthropics/claude-for-legal/blob/main/ai-governance-legal/skills/reg-gap-analysis/SKILL.md>

Building Responsible Agentic AI Systems

The Summary

- LLM is ANOTHER kind of intelligence.
- It answers every question!
- However, it may answer with a plausible but incorrect response (it only predict the next word.)
- This is a structural problem in LLM.
- We must be very discerning.
- Good news: we can guide/navigate the conversation with LLM to align with our logic (e.g., value, principles, implementation)
- So LLM can manifest the trajectory of its decision-making to us.
- That's today's AI agents!